Brunhes Epoch Geomagnetic Secular Variation on Terceira Island, Central North Atlantic

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A paleomagnetic survey has been made of Brunhes epoch lavas and ignimbrites on Terceira in the Azores. At least six widely spaced samples were taken from each of 33 separate bodies. The corrected dispersion of virtual geomagnetic poles (S_F) is 13.9 (with upper and lower 95% confidence limits of 16.3 and 12.2, respectively), consistent with the present geomagnetic field behavior in the same latitudes. Analysis of the data as nine independent surveys involving from two to seven separate specimens per body provides values of S_F that are not significantly different. Thus an equally accurate value of S_F could have been obtained with less than half the samples collected. Analysis of the effect of K_W (the within-body precision parameter) and other factors involved in the computation of S_F suggests that the largest number of separate cores per separate body required for defining S_F is four. Two separate lavas recorded a possible single departure of the geomagnetic field, and thus evidence of instability of the geocentric dipole during the Brunhes epoch is provided.

Paleomagnetic surveys of Brunhes epoch basalts have facilitated defining the geomagnetic secular variation for the past 0.7 m.y. Compilation of such data over a range of latitudes can also be used to test conventional geomagnetic field models, all of which involve various relative contributions of the main centered dipole and nondipole components to the total field.

In model A, vectors of varying orientation but constant amplitude (to simulate nondipole activity) are added to a fixed axial dipole [*Irving and Ward*, 1964]. In model B the geomagnetic field variation is attributed entirely to wobble of the main dipole [*Creer et al.*, 1959]. Model C, largely developed from the analyses of *Cox* [1962] and *Creer* [1962], combines dipole wobble and a randomly varying nondipole component. For the Brunhes the total secular variation can be calculated by the equation

$$S_p = (S_D^2 + S_N^2)^{1/2} \qquad (1)$$

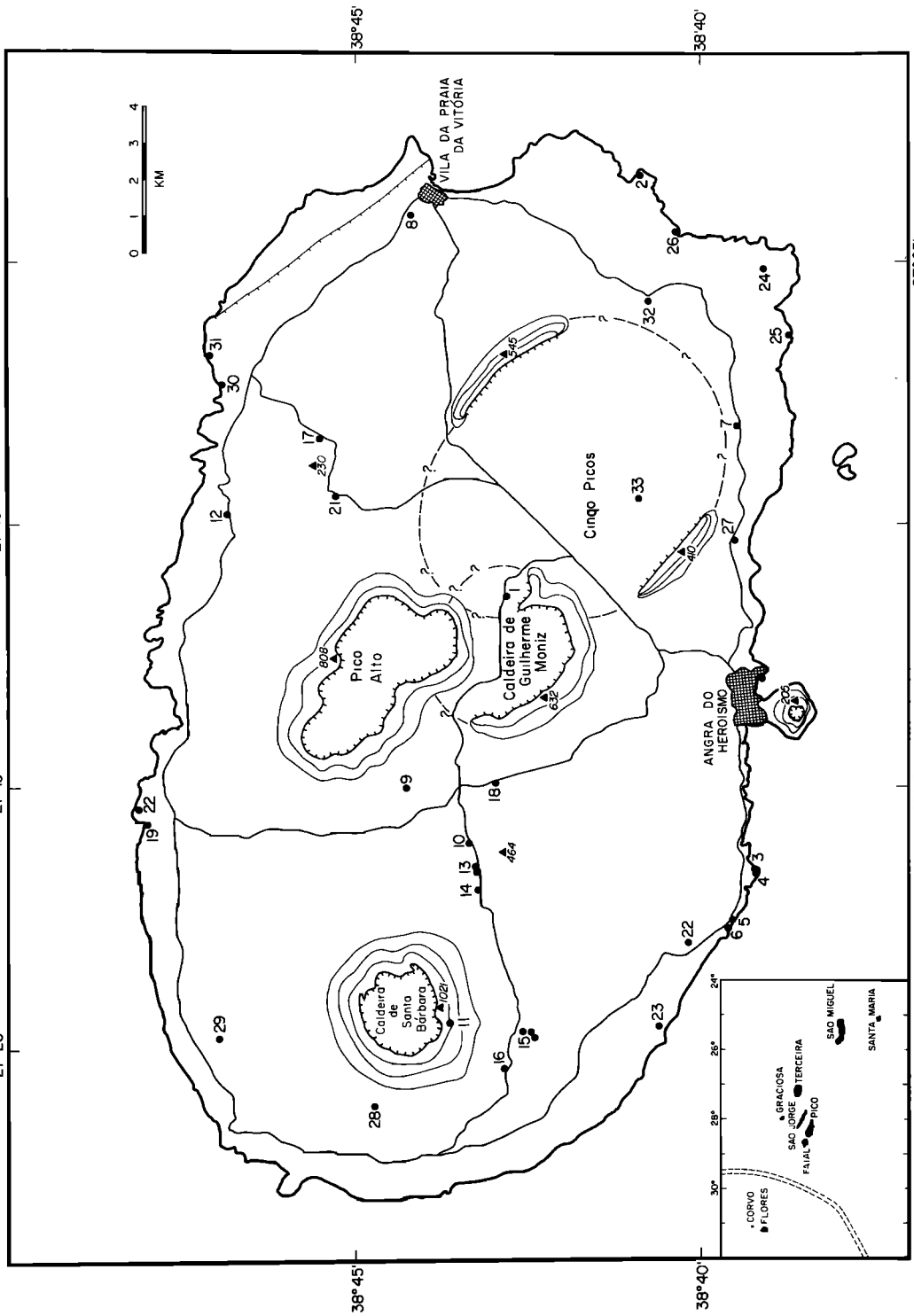
where S_D is the maximum dipole wobble, which *Doell* [1969] has shown to be 11½°, and S_N is the present nondipole behavior, obtained by combining the virtual geomagnetic pole (VGP) for the nondipole field at 10° longitude intervals, averaged between hemispheres.

Cox and Doell [1964] have shown that the most convenient method of representing these models for testing using paleomagnetic data is by expressing them as site latitude versus angular standard deviation S_T of the VGP for each of B separate bodies (or points in time), where

$$S_T^2 = \frac{1}{(B-1)} \sum_{i=1}^{B} \delta i^2 \qquad (2)$$

and δi is the angular separation between the ith lava VGP and the reference pole, which is usually the mean VGP. In the Brunhes epoch this mean VGP should approximate the present geographic pole. In (2), S_T represents the total measured dispersion, which includes contributions from experimental and natural sources, in addition to the geomagnetic field variations that are ideally isolated for analysis. As summarized by *Cox* [1969a] and *Doell* [1970], one method of correction for these factors is the application of the following equation:

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$$(1/K_F) = (1/K_T) - (1/\bar{N}K_W) - (1/K_A) \quad (3)$$

where K is the precision parameter, the subscript F refers to the ambient ancient geomagnetic field directions, the subscript T refers to the total measurement, which includes laboratory and field contributions, the subscript W refers to the within-lava precision, and the subscript A refers to the effect of the magnetic anomaly existing during extrusion of the lava on the precision of the remanent magnetism recording of the ambient geomagnetic field direction. The average number of separate cores per lava is given by $\bar{N}$.

The best estimate of true ancient geomagnetic field dispersion S_F can be then related to K_F by

$$S_F^2 \approx 2/K_F \quad (4)$$

where S_F is in radians. In this way, S_F can be obtained.

The within-lava precision parameter [*Watson and Irving*, 1957] is obtained from

$$K_W = \sum_{i=1}^{N} \frac{(\bar{N}_i - 1)}{(\bar{N}_i - R_i)} \quad (5)$$

where there are a total of N specimens for all lavas combined, each lava contributing $\bar{N}$ unit vectors that have a resultant vector length of R.

The value of K_A depends on the intensity of magnetization J of the lava involved. *Doell* [1970], using the methods of *Cox* [1970], has estimated that, given widely separated specimens, a value of $K_A = 2100$ is appropriate for lavas with $J \approx 10^{-3}$ emu cm^{-3}.

To facilitate comparison of S_F with the various geomagnetic field models described earlier, it is most convenient to compute the precision parameters given in (2)–(5) by analyzing the VGP (rather than directions) for each core. A critical part of the analytical method is data rejection. Although the exclusion of data from lavas with VGP latitudes less than 45° would appear a reasonable practice (to discriminate against geomagnetic behavior involving nonaxial dipole activities or 'departures') and apparently has a physical basis [*Wilson et al.*, 1972, Figure 5], the exclusion of data from lavas with 95% confidence ovals greater than 9°, for example, appears somewhat arbitrary in our opinion.

Doell and Cox [1963] have argued that, given the best possible field and laboratory precision, the optimum number of cores required to adequately define K_W (and therefore K_F and S_F) is from six to eight per lava. Above this sampling density, little is to be gained. There has been some suspicion that S_F may be satisfactorily determined with a lower number of separate specimens [*Stone*, 1970], but this has not been demonstrated. In fact, *Doell and Cox* [1971] specifically point out that their studies using the above sampling density for Brunhes epoch lavas have not been used by other workers, so that comparisons cannot be made between results from the geographic regions represented by their studies and those made elsewhere. *Cox* [1969a] has, for example, rejected *Chamalaun's* [1968] result for Reunion Island, which used an average of less than two separate specimens per lava, so that K_W (and hence S_F) could clearly not be derived.

This paper presents the results of a study of Brunhes epoch paleomagnetic secular variation for the central North Atlantic made by analyzing the remanent magnetism of lavas for Terceira Island in the Azores (Figure 1). In addition to contributing to testing of the geomagnetic field models for the Atlantic region, the data will be analyzed to answer the following question: would the measurement of S_F (which is based on an average of more than seven widely separated specimens per body) have been significantly different if we had terminated the sampling of each lava (or point in time) after only 6, 5, 4, 3, or even 2 cores had been taken? We shall also use our results to estimate how critical the estimate of K_A and the measurement of K_W are to our final result.

GEOLOGY OF TERCEIRA ISLAND

Ridley et al. [1973] have presented a comprehensive summary of the evolution, volcanology, geochemistry, and geophysics of the Azores

Fig. 1. (Opposite) Sketch map of Terceira Island, showing location of the 33 sample sites. Major towns and roads shown. The four major volcanic centers are named. Probable old limits of the two southwestern volcanic centers are added. Insert shows location of Azores islands and position of mid-Atlantic ridge. The line of the Terceira rift (a secondary spreading center) is not shown but is along the line Graciosa–Terceira–Sao Miguel [*Krause and Watkins*, 1970, Figure 12]. Heights of local peaks (solid triangles) given in meters.

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region. Other summaries of the volcanism of the Azores are by *Machado* [1965], *Machado et al.* [1972], and *Zbyszewski* [1963]. The common tephra and ignimbrites on San Miguel and Terceira have been described by *Walker and Croasdale* [1971] and *Self* [1971].

Krause and Watkins [1970] have postulated a model for the evolution of the islands suggesting that the Terceira rift (Figure 1) is a secondary spreading center initiated as a result of a change in the direction of crustal spreading accompanying a change in trend of the mid-Atlantic ridge from north-south to its present configuration (Figure 1) during the early Tertiary.

The Terceira rift is expressed on the island of Terceira (Figure 1) as northwest-southeast–trending fissures on which are superimposed the four stratovolcanoes constituting most of the island. Only Pico Alto lies off the visible fissures. The fissure zone is also marked by a line of basaltic cinder cones, spatter cones, lava eruptions, and trachyte and rhyolite domes. The two volcanoes in the southeast of the island are the oldest and are extinct. The other two volcanoes are younger and not extinct. The most recent eruption (in 1761) was directly from the fissure zone, producing alkali-olivine basalt lavas.

Pico Alto grew to its present altitude mainly as the result of explosive pumice-producing eruptions and lava flows of trachytic composition with only rare basaltic material. A caldera formed, probably in association with at least two identifiable ignimbritic eruptions.

Cinquo Picos is a caldera 7 km in diameter with basaltic and trachytic lavas exposed in the rim of the caldera. These lavas are suspected to be the oldest on the island. On the flanks of the caldera are scattered parasitic basaltic eruptions.

Guilherme Moniz is a steep-sided circular caldera, the northern part of which is swamped by trachyte flows from the higher-level Pico Alto volcano. The older parts of the volcano are basaltic. Their extrusion was followed by acidic pyroclastic and lava eruption and caldera collapse. It appears probable that Pico Alto grew as a large acidic parasitic dome on the northern slope of Guilherme Moniz volcano.

Santa Barbara developed initially as a basaltic ash and lava stratovolcano and changed later toward acidic eruptions, which included late-stage rhyolitic pumice falls associated with the younger of the later two calderas. Basaltic adventive cones and a series of trachyte domes and flows are associated with the main fissure trend.

No paleontological age estimates are available for the island, and isotopic dating has been confined to the ^{14}C method, which provides ages between 18,000 and 23,000 years for selected eruptions. On geomorphological grounds, however, there can be little doubt that periods of the order of hundreds of thousands of years are represented. As we shall show, the island is overwhelmingly confined to the Brunhes epoch (t = 0–0.69 m.y.) in outcrop and is therefore ideally suited for our experimental purposes as described earlier.

Serughetti and Roche [1968] have already presented Brunhes epoch paleomagnetic data from 22 lavas on Flores (Figure 1 inset), which *Watkins et al.* [1972] have used to derive a value of $S_T = 15.7$, upper and lower 95% confidence limits [*Cox*, 1969*b*] being 19.8 and 13.0, respectively. No value of S_F could be computed (equations 3 and 4) since no data were presented for individual specimens, although an average of five samples was collected from each lava. *Saucier and Roche* [1964] have also presented some paleomagnetic results (based on an unspecified number of specimens) from 14 lavas on Faial and seven bodies on Graciosa (Figure 1). No paleomagnetic results from Terceira have previously been published.

METHODS

Field. A total of 236 separate cores 2.5 cm in diameter have been taken from 33 separate lavas and ignimbrite flows on the island by a gasoline-powered portable drill [*Doell and Cox*, 1965]. Orientation of each core was made with a sun compass to a precision of ±2.5°, and cores were separated laterally by an average of greater than 10 meters. Polarity measurements using a compass supplemented field observations to restrict all sampled units to the Brunhes epoch.

The location of each body sampled is given in Figure 1 and also in the appendix, to which is added a petrological description of each sampled unit. The number of separate cores in each body is included in Table 1.

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TABLE 1. Paleomagnetic Data for Terceira Island

Lava*	N	D, E°N	I, deg	R	J, emu $g^{-1} \times 10^{-4}$	K	α_{95}, deg	ϕ', °E	θ', °N
				Lavas with Low-Latitude VGP					
8	6	172.3	18.7	5.8556	1.42	34.6	11.5	163.0	41.1
23	7	179.8	40.3	6.9859	1.75	425.1	2.9	153.0	28.4
				Analysis of Directions					
1	7	11.5	64.8	6.4505	3.18	10.9	19.1	15.3	78.3
2	6	356.6	52.0	5.9932	3.27	736.0	2.5	178.1	83.4
3	6	342.7	38.1	5.5986	1.07	12.5	19.8	198.4	67.2
4	7	2.2	57.7	6.8803	0.83	50.1	8.6	73.4	88.2
5	8	357.7	17.3	7.9778	0.76	315.6	3.1	157.4	60.1
6	6	8.0	58.6	5.6923	0.42	16.3	17.1	54.5	83.8
7	8	354.5	41.2	7.8114	1.62	37.1	9.2	171.7	74.2
9	6	348.8	72.4	5.9531	3.79	106.6	6.5	315.3	69.7
10	7	353.1	60.2	6.9738	2.51	229.4	4.0	269.9	84.2
11	7	15.7	50.3	6.5833	0.93	14.4	16.5	88.8	75.1
12	7	326.8	54.6	6.9851	1.55	404.0	3.0	245.5	63.4
13	7	346.1	56.8	6.9950	3.30	1211.0	1.7	240.0	79.0
14	7	350.6	59.6	6.9888	2.58	535.9	2.6	259.3	82.6
15	7	0.4	51.5	6.9880	0.93	501.1	2.7	149.5	83.5
16	8	347.7	49.0	6.9159	1.12	6.5	23.6	205.4	76.6
17	7	335.4	53.2	6.9415	1.81	102.6	6.0	236.3	69.6
18	7	21.4	54.8	6.9630	1.62	162.2	4.8	67.1	72.7
19	8	357.5	51.2	7.9359	2.00	109.3	5.3	170.2	82.9
20	7	5.7	59.3	6.7198	2.37	21.4	13.3	43.1	85.4
21	8	331.2	53.3	7.9771	1.77	306.0	3.2	239.9	66.4
22	8	3.5	44.4	7.3617	0.79	11.0	17.5	138.6	77.1
24	6	9.7	52.1	5.9737	1.00	190.2	4.9	97.0	80.1
25	8	38.9	57.2	7.9911	1.56	786.4	2.0	52.1	59.6
26	8	356.6	50.9	7.9785	2.77	326.3	3.1	175.0	82.4
27	8	339.5	62.0	3.3734	0.56	1.5	74.5	265.7	74.0
28	8	9.9	25.0	7.9784	2.18	323.8	3.1	131.0	63.0
29	6	11.8	56.1	5.9657	1.04	146.0	5.6	71.5	80.5
30	8	343.4	51.1	7.9227	2.14	90.6	5.9	220.7	74.8
31	8	7.0	54.8	7.9707	2.13	239.3	3.6	91.5	83.5
32	6	6.8	57.0	5.0044	1.28	5.0	33.2	72.8	84.5
33	8	2.5	33.0	7.3706	3.47	11.1	17.4	146.0	69.2
				Analysis of VGP					
1				5.8442		5.2	29.3	15.8	78.5
2				5.9904		520.3	2.9	178.5	83.4
3				5.7124		17.4	16.5	201.5	68.6
4				6.7916		28.8	11.4	49.4	88.6
5				7.9848		461.6	2.6	157.4	60.1
6				5.7227		18.0	16.2	37.9	81.3
7				7.7438		27.3	10.8	172.5	74.2
9				5.8757		40.2	10.7	315.7	69.3
10				6.9519		124.8	5.4	271.7	84.1
11				6.2825		8.4	22.2	87.5	75.0
12				6.9752		241.9	3.9	245.7	63.3
13				6.9920		751.1	2.2	240.2	79.0
14				6.9813		321.2	3.4	260.1	82.6
15				6.9799		298.3	3.5	149.4	83.5
16				6.7369		5.5	25.9	220.1	76.9
17				6.8916		55.4	8.2	236.6	69.5
18				6.9501		120.3	5.5	66.2	72.7
19				7.9183		85.6	6.0	170.3	83.2
20				6.4062		10.1	20.0	37.0	84.5
21				7.9635		191.7	4.0	240.1	66.4
22				7.0960		7.7	21.3	138.2	77.4
24				5.9577		118.3	6.2	96.5	80.1
25				7.9866		522.2	2.4	52.0	59.7
26				7.9750		280.5	3.3	175.7	82.5
27				2.8483		1.4	88.1	268.8	74.2
28				7.9844		449.8	2.6	131.0	63.0
29				5.9342		75.9	7.7	71.4	80.7

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TABLE 1. (continued)

Lava*	N	D, E°N	I, deg	R	J, emu $g^{-1} \times 10^{-4}$	K	α_{95}, deg	ϕ', °E	θ', °N
30				7.9098		77.6	6.3	222.2	74.7
31				7.9563		160.1	4.4	90.5	83.6
32				4.3530		3.0	46.4	48.9	86.4
33				7.2290		9.1	19.4	146.3	69.2

N, number of separate cores; D and I, declination and inclination of remanent magnetism after treatment in alternating magnetic fields, respectively (I is in degrees below the horizontal); R, resultant vector, applying unit vector per specimen; J, intensity of magnetization; $K = (N - 1)/(N - R)$; α_{95}, semivertical angle of 95% confidence cone [*Fisher*, 1953]; ϕ' and θ', longitude and latitude of VGP, respectively.

*Flow number (see Figure 1 and appendix).

Laboratory. The remanent magnetism of a specimen 2.2 cm in length from each core was determined by a 5-Hz spinner magnetometer. Unstable components were minimized by applying alternating magnetic fields of 100 and 200 oe to each specimen, the remeasurement of the remanent magnetism following each treatment.

Computations. Mean directions of remanent magnetism were computed by a minimum dispersion criterion (see *Watkins et al.* [1972] for discussion). All data were retained for computation of S_F by (2)–(5), except for those bodies with VGP equatorward of the 45° latitude, since these clearly represent anomalous behavior of the geomagnetic field not relevant to the study of geomagnetic secular variation [*Doell and Cox*, 1972]. No other data selection has been made.

RESULTS

Departures. Two of the lavas provide results with low-latitude VGP. These are presented in Table 1.

All results. Table 1 presents the mean remanent magnetism directions and associated confidence parameters for each body with a VGP latitude greater than 45°. Table 1 also presents the same data for each unit in the form of the results of analyses of the VGP for each specimen.

Selected groups. Table 2 presents results that would materialize from a series of more limited sampling systems. For all bodies (all of which have from six to eight separate cores) we in effect cease sampling after the first six cores, the first five cores, and the first four cores. We also compile essentially independent results from the first and second series of three cores and from three separate series of two cores. By this method we therefore arrive at an additional eight independent data sets, each with a consistent sample spacing. (Random compiling of core sets from each body could lead to widely differing between-core spacing.)

Computation of S. Table 3 presents all computations of the data from Table 1 to field values of S_F (equations 2–5). These computations are done with respect to both the present spin axis and the mean pole position. The 95% confidence intervals were calculated by using the tables of *Cox* [1969*b*].

TABLE 2. Analysis of Groups with Varying Numbers of Cores per Body (VGP)

Group*	$\bar{N}$†	B	R	K	α_{95}	ϕ', °E	θ', °N
1	7.2	31	29.8510	26.1	5.2	170.6	85.6
2	6	31	29.7642	24.3	5 4	170.5	84.4
3	5	31	29.6489	22.2	5.6	175.6	84.7
4	4	31	29.5329	20.4	5.9	175.5	83.0
5	3	31	29.4290	19.1	6.1	177.5	86.3
6	3	30	28.6395	21.3	5.8	167.4	84.2
7	2	31	29.1973	16.6	6.5	170.9	85.0
8	2	28	26.4712	17.7	6.7	173.4	85.6
9	2	30	28.7632	23.4	5.5	167.1	84.6

B, number of separate lavas (note that B is not constant because of the rejection of data for bodies with $\theta' \leq 45°$); R, resultant vector, applying unit vector per specimen; $K = (N - 1)/(N - R)$; α_{95}, semivertical angle of 95% confidence cone [*Fisher*, 1953]; ϕ' and θ', longitude and latitude of VGP, respectively.

*Number refers to the separate survey extracted from the data. Group 1 is the entire survey.

†Average number of cores per flow. Other values of $\bar{N}$ are first 6, 5, 4 cores collected, respectively. Two values of $\bar{N}$ result from taking the first three and second three cores from each body as separate groups. The three groups with $\bar{N} = 2$ are from the first, second, and third pairs of cores for each body.

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DISCUSSION

Departures. Lavas 8 and 23 (Figure 1) have recorded VGP in the vicinity of 28°–40°N, 160°E (Table 1). *Doell and Cox* [1972] have presented paleomagnetic data from over 1000 separate upper Tertiary lavas from various parts of the world showing that departures of VGP from the present geographic pole greater than 50° of latitude are recorded by less than 2% of the sampled units. The phenomenon is clearly of short duration, perhaps less than 1000 years, and can result in principle either from nondipole activity during periods of nondipole intensity or from a rapid tilting of the main dipole. The similarity of the virtual geomagnetic longitudes of lavas 8 and 23 (Table 1) suggests that they may have recorded the same departure in corresponding sequence, if the departure was due to tilting of the main dipole, since otherwise it is unlikely that the VGP longitudes would be so similar.

This very unusual direction of remanent magnetism should provide a powerful stratigraphic marker on Terceira. The result is to be added to data from Hawaii [*Doell and Cox,* 1972] and the southern Indian Ocean [*Watkins and Nougier,* 1973], which show that periods of instability of the geocentric dipole may have occurred several times during the Brunhes epoch.

Secular variation in the Atlantic region. Figure 2 shows the value of S_F for the Terceira lavas and ignimbrites, together with S_F-latitude relationships for the various geomagnetic field models mentioned earlier. We have added the independently published determinations of S_F for Iceland [*Doell,* 1972] and France [*Doell,* 1970]. The value of S_F for the Canary Islands has been recomputed from the data of *Watkins et al.* [1972], which originally involved presentations of S_T only, since all available data for the regional comparisons made in that presentation were insufficient for computation of S_F (Table 4).

The measurements of S_T and S_F for the North Atlantic confirm the earlier results of *Watkins et al.* [1972], which show that the dipole wobble model is to be excluded from further consideration during the Brunhes epoch. In addition, no distinction can be drawn between the other two models.

Effect of varying sampling systems. Figure 3 illustrates the results of our consideration of different sampling systems, which are summa-

TABLE 3. Computation of Dispersion of VGP for Varying Numbers of Cores per Body

Group*	$\bar{N}$	K_W	K_T	K_F	S_T	S_F	S_u	S_l
			With Respect to Geographic Pole					
1	7.2	15.30	23.67	30.64	16.7	14.6	17.2	12.8
2	6	16.65	21.46	27.68	17.5	15.4	18.4	13.2
3	5	18.52	19.99	25.81	18.1	15.9	19.1	13.7
4	4	27.81	17.50	20.98	19.4	17.7	21.1	15.2
5	3	17.55	18.13	28.03	19.0	15.3	18.3	13.2
6	3	27.86	18.96	24.81	18.6	16.3	19.5	14.0
7	2	22.95	15.36	23.33	20.7	16.8	20.1	14.4
8	2	20.08	16.47	28.31	20.0	15.2	18.4	13.0
9	2	38.16	20.92	29.23	17.7	15.0	18.0	12.9
			With Respect to Mean Pole					
1	7.2	15.30	25.72	34.16	16.0	13.9	16.3	12.2
2	6	16.65	23.96	32.01	16.6	14.3	17.1	12.3
3	5	18.52	21.92	29.11	17.3	15.0	18.0	12.9
4	4	27.81	20.13	24.87	18.1	16.2	19.4	14.0
5	3	17.55	18.83	29.72	18.7	14.9	17.8	12.8
6	3	27.86	21.00	28.43	17.7	15.2	18.2	13.0
7	2	22.95	16.27	25.50	20.1	16.0	19.2	13.8
8	2	20.08	17.29	30.80	19.5	14.6	17.6	12.5
9	2	38.16	23.07	33.60	16.9	14.0	16.8	12.0

$\bar{N}$, average number of cores per flow; K_W, within-body precision parameter; K_T, total precision parameter; K_F, corrected precision parameter; S_T, total dispersion; S_F, corrected dispersion; S_u and S_l, upper and lower 95% confidence parameters for S_F, respectively.
*Number refers to the separate survey extracted from the data. Group 1 is entire survey.

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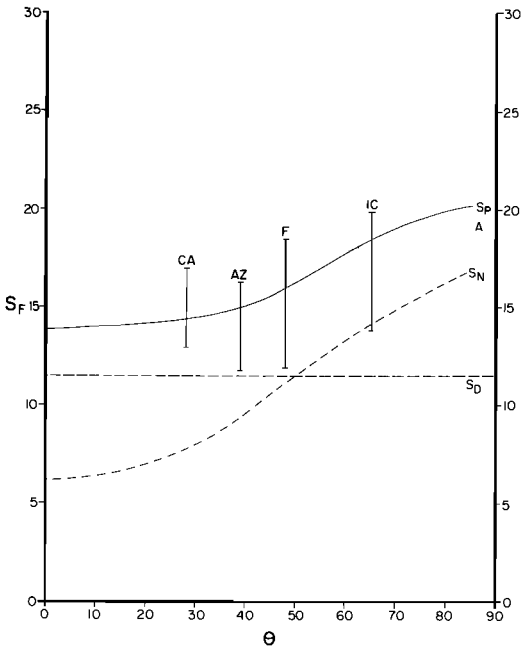


Fig. 2. Variation of corrected dispersion of VGP (S_F) as a function of latitude (θ) for the Atlantic region. Results given in the form of 95% confidence limits for S_F. Here AZ indicates Azores (this paper); CA, Canary Islands; F, France; IC, Iceland. See Table 4 for detailed results and data sources. See text for origin of geomagnetic field models associated with the S_P, A, S_N, and S_D versus θ relationships shown.

rized in Table 3. We have in effect created an additional eight separate collections from our set of samples, which vary from two to six separate cores per body. All results are for S_F (including corrections for K_W and K_A, as given in (2)–(4)). The only data rejection applied is exactly the same as that for the entire collection result: units yielding VGP latitudes lower than 45° in the recomputed results have been excluded. As Figure 3 clearly shows, although small differences in the 95% confidence limits for S_F emerge, no significant difference is seen in the measured value of S_F. This can only be interpreted to indicate that nothing significant has been gained in defining S_F in this particular study by sampling more than three cores per body. We cannot be certain, however, that the same conclusion could be applied to other studies.

According to our result (Figure 3), the practice of taking at least six cores per unit for paleomagnetic secular variation studies may not be necessary. As Table 3 clearly shows, the mean VGP position varies hardly at all with variation of the sampling system. In fact, the difference between S_T and S_F for the entire collection is not only slight (only 15%) but also does not vary greatly when $\bar{N}$ varies between 3 and 7. The value of S_T that *Watkins et al.* [1972] derived from the data of *Serughetti and Roche* [1968] for 22 lavas from Flores (Figure 1) is 15.7 with upper and lower 95% confidence limits set at 19.8 and 13.0, respectively, which compare well with the upper and lower confidence limits for our collection from Terceira (Figure 3). We consider below the possible effect of other factors on the precision of S_F.

Effect of K_A variation. As *Cox* [1970] and *Doell* [1970] have discussed, calculation of the distorting effect of the preexisting magnetic anomaly on the remanent magnetism of a cooling lava is subject to uncertainties. In their original publication on the evaluation of such effects, *Cox and Doell* [1964] used the only suitable magnetic anomaly map of a basaltic island available at that time [*Rikitake et al.*, 1951; *Yokoyama*, 1957] to compute the distorting effect. Inclination anomalies as large as $3\frac{1}{2}°$ and several kilometers long are found on Ooshima

TABLE 4. Dispersion of VGP for Other Atlantic Region Studies

Area	N	K_W	K_T	K_F	S_T	S_F	S_u	S_l	Reference
France	31	55.1	27.9	30.8	15.7	14.8	18.4	12.9	*Doell* [1970]
Iceland	28	77.9	24.1	25.9	16.6	16.2	19.8	13.7	*Doell* [1972]
Canary Islands	45	22.8	23.6	30.1	16.7	14.8	17.1	13.0	*Watkins et al.* [1972]

N, average number of cores per flow; K_W, within-body precision parameter; K_T, total precision parameter; K_F, corrected precision parameter; S_T, total dispersion; S_F, corrected dispersion; S_u and S_l, upper and lower 95% confidence parameters for S_F, respectively.

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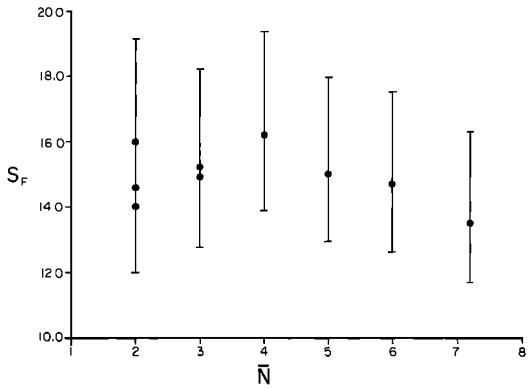


Fig. 3. Variation of corrected dispersion of VGP (S_F) as a function of the number of specimens per body ($\bar{N}$) with 95% confidence limits added. For $\bar{N} = 2$ and $\bar{N} = 3$, there is more than one determination (Table 4): all values of S_F are presented, and the extreme 95% confidence limits are added. See text and Table 4 for specimens used to compute S_F groups.

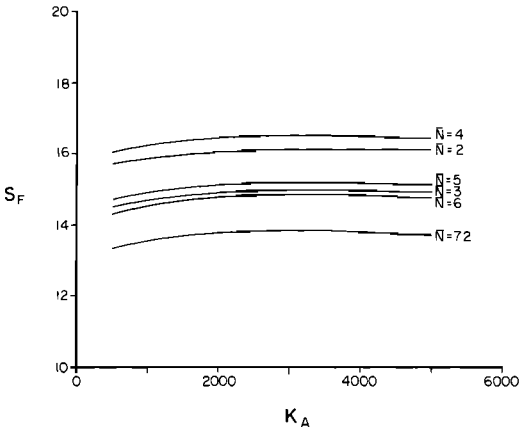


Fig. 4. Variation of corrected dispersion of VGP (S_F) as a function of the precision value K_A (an expression of the lack of parallelism of the remanent magnetism direction and the regional geomagnetic field direction caused by the influence of the magnetic anomaly existing prior to extrusion of the selected series of flows).

Island. The value of K_A can then be computed by determining on a grid the difference between the regional geomagnetic field direction and that observed locally on the island (and therefore to be expected in a hypothetical paleomagnetic recording). The best present estimate of K_A is considered to be 2100 by *Doell* [1970], depending on the intensity of magnetization of the lavas involved. A decrease in K_A from 5000 to 500 will decrease our computed value of S_F from 13.7 to only 13.3. Particularly when the 95% confidence parameters are considered, the significance of the exact value of K_A would seem to be minor, as was previously pointed out by *Doell* [1972]. The same conclusion applies to our 'additional' eight surveys: as Figure 4 shows, S_F is (for the Terceira Island survey) virtually unaffected by varying K_A, even for $\bar{N} = 2$–6.

Effect of K_W variation. Figure 5 shows, for the Terceira Island data, the effect on S_F of varying K_W for various values of $\bar{N}$. Values of K below 20 can be seen to have a substantial effect on computations of S_F for the Terceira collection, particularly for $\bar{N} = 2$ and $\bar{N} = 3$.

It is important to realize that, for our collection, K_W tends to decrease with increasing $\bar{N}$ (Table 3). The value of K_W depends on within-lava effects, such as postcooling block rotation and the effect of preexisting magnetic anomalies as discussed earlier, in addition to field orientation precision and the inherent limitation or capacity of the remanent magnetism of a cylindrical lava or ignimbrite specimen 2.5 cm in diameter and 2.2 cm long to integrate out to the precise direction of the local ambient magnetic field. On the basis of our results from Terceira, it would appear that seven specimens spaced at regular intervals of about 10 meters are more likely to record faithfully the upper limit of the natural contributors to K_W than are two to four or five specimens at the same spacing.

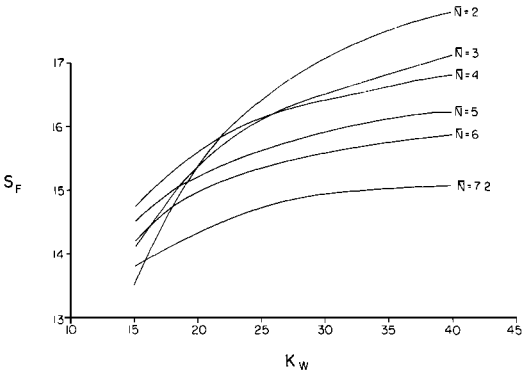


Fig. 5. Variation of corrected dispersion of VGP (S_F) as a function of the within-lava precision parameter (K_W) for various values of the number of specimens per lava ($\bar{N}$). The curves for $\bar{N} = 2$ and $\bar{N} = 3$ are presented for the first two and three specimens collected from each body, respectively.

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But, at the same time, K_T tends to be less for values of $\bar{N}$ lower than 6 (Table 3), and thus the increased K_W values are offset in providing any substantial variations in K_F (Figure 3).

CONCLUSIONS

Brunhes epoch secular variation in the North Atlantic is similar to that of the present geomagnetic field. The dipole wobble geomagnetic field model does not fit the data.

Our discussion of the importance of K_W to the determination of K_F serves to emphasize the importance of data rejection for any sampled body. If the data from one core are rejected (and thus provide a higher K_W for a given unit) and if this selectivity is applied to several bodies, a clearly higher value of S_F can be expected, unless the value of K_W for the body is above 40 or so before application of any rejection criterion.

In our study of Terceira Island we have not improved our determination of S_F by sampling more than three cores in each body. Our consideration of the effect of K_W on the measurement of S_F suggests, however, that four is the best number of cores per body required for satisfactory determination of S_F. An accurate value of K_A is not important in determining S_F. Although this approach to the problem of the optimum number of separate specimens required for satisfactory definition of S_F is essentially empirical, in contrast with the comprehensive field and laboratory methods analyses of *Cox and Doell* [1964] that produced the optimum number of six to eight cores per lava, which they have since adhered to [*Doell and Cox*, 1972], we suggest that analyses of other similar surveys may be profitably made to evaluate the validity of our preferred simpler sampling system.

APPENDIX: SITE AND PETROLOGICAL DESCRIPTION OF EACH BODY SAMPLED

AT-01 Olivine-augite basalt, dated at 2110 ± 120 years by ^{14}C (unpublished data). Flowing across flat caldera floor in an easterly direction. Flow thickness at sampled point is 3 meters.

AT-02 Porphyritic olivine basalt with feldspar phenocrysts. Thickness of flow approximately 5 meters on lava delta where samples were taken.

AT-03 Vesicular basalt lava with scattered fine plagioclase phenocrysts. An old aa flow exposed only in the base of the cliffs but possibly derived from the fissure zone, giving a south flow direction on an unknown but possibly small slope.

AT-04 Welded ignimbrite, gray to yellow, trachytic composition. Composed of welded crystals, black glassy fragments, small lithic fragments, and ground mass in a eutaxitic texture. Flow direction from northeast on a low slope of 5°. Age approximately 18,600 years according to ^{14}C (University of Birmingham, England, Dating Laboratory, 1973).

AT-05 Thick flow of olivine-augite basalt, very massive flowing to south coast. Sampled in cliffs at edge of flow. A young lava, possibly about 10,000 years, and originating from a cinder cone 4 km to the north. Flow direction south on low slopes of less than 3°.

AT-06 Same site as AT-05 but an older feldspar-rich basaltic lava with a few augite crystals. Up to 10 meters thick, aa, with many flow structures. Originating possibly from the fissure zone or from the proto Santa Barbara volcano.

AT-07 Basalt lava approximately 3 meters thick, originating from one of two prominent cones to the north, near the edge of the old Cinquo Picos caldera. A vesicular olivine-augite basalt, typical of the postcaldera lavas of the oldest volcano. Flow direction south on a 2°–3° slope.

AT-08 Vesicular massive basalt flow consisting of at least three flow units of aa lava. An old flow, associated with the volcano-building stage of Cinquo Picos volcano and therefore representative of the early island lavas. No slope, actual flow direction unknown but probably north-northeast.

AT-09 1761 olivine-augite basalt flow. Sampled at a place where the lava was ponded, therefore effectively no flow direction or slope. One aa flow unit about 2–3 meters thick.

AT-10 A recent fissure lava sampled near the fissure line. Two flow units of highly vesicular lava that must have been very

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fluid. A poorly porphyritic olivine basalt pahoehoe-type flow with very large gas bubbles in the lava. Flow direction south to southeast on a slope of about 5°.

AT-11 One of the more recent Santa Barbara basaltic lavas from the main vent of the volcano. Sampled near top of volcano. A highly fluid, often spongy and very vesiculated aa-type lava. Flow toward south on slope of 15°–20°.

AT-12 A porphyritic olivine-feldspar basalt consisting of two flow units and about 1.5 meters thick. This vesicular aa flow came from an early parasitic cone of Pico Alto volcano and flowed north to northeast on a slope of about 5°.

AT-13 Another recent fissure lava adjacent to AT-10. The site and slope are similar, but this is a slightly younger lava. A porphyritic aa flow with small feldspar laths and some olivine, about 5 meters thick.

AT-14 Another lava with origin and site similar to those of AT-10 and AT-13. This is a feldspar-olivine pahoehoe flow, one of a series of small lava flows erupted from the fissure, mainly flowing south.

AT-15 A highly porphyritic feldspar basalt with large plagioclase crystals and some spongy augite. One of the volcano-building lavas of Santa Barbara, of very widespread occurrence on the south-southwest side of the volcano. Flow direction to the south on a 10°–12° slope at sample point.

AT-16 Another Santa Barbara basalt flow, younger than AT-15. A poorly vesicular feldspar lava with signs of great fluidity, including stretched vesicles. Flow toward southwest on a 40° slope.

AT-17 An old olivine-augite basalt, flow about 2–3 meters thick, flowing toward the north-northeast on a low slope of 3°–5°.

AT-18 Vesicular basalt from one of the cones on the fissure in the center of the island. Flowing south to southwest on a negligible slope.

AT-19 A feldspar-phyric basalt lava forming an old lava delta on the north coast of the island. Flow in a northerly direction on a low slope of less than 3° at point of sampling.

AT-20 Same locality as AT-19 but a younger lava delta extending further north. An olivine-augite basalt with scattered olivine nodules and crystal clots. Separated from AT-19 by two basaltic ashes and an ignimbrite.

AT-21 A porphyritic olivine basalt from small adventive cone on lower flanks of Pico Alto. An old flow, dating from before the explosive trachytic phase of Pico Alto. Flowing toward north-northeast on a slope of 5°.

AT-22 Old fissure zone or proto Santa Barbara basalt flowing south toward the sea on a slope of 3°. A very fluid, vesicular aa flow.

AT-23 A lava similar to AT-22, probably from early Santa Barbara site on coast at southernmost exposure of flow.

AT-24 Vesicular, poorly porphyritic basalt (with some small feldspar crystals). A thickness of 1–1.5 meters, flowed southwest on a 10° slope.

AT-25 Very vesicular olivine basalt, approximately 15 meters thick and comprised of four to five flow units. Both pahoehoe and aa structures observed in this compound flow. Top unit was sampled.

AT-26 Massive olivine basalt, 3 meters thick with nodules included within the basalt. Flow was eastward over a 2°–3° slope.

AT-27 Old feldspar porphyritic lava, possibly a trachyte. Lava is a massive light gray, 5 meters thick.

AT-28 Fine-grained slightly vesicular basalt of unknown thickness, sampled in a stream bed.

AT-29 Moderately porphyritic (feldspar and minor olivine) vesicular basalt of unknown thickness.

AT-30 Poorly porphyritic olivine basalt with small augites. A fluid aa flow, 5 meters thick and flowing toward the north, forming a delta at sampling site.

AT-31 Massive olivine-feldspar lava but vesicular where sampled. Two flow units with total thickness of approximately 5 meters.

AT-32 Feldspar and augite trachy-basalt, massive with unknown thickness.

AT-33 Vesicular, porphyritic olivine-feldspar-augite basalt composed of two units with a total thickness of 4 meters.

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